

# Erfan Amani Bani

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## Education

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| <b>London Business School</b><br><i>Ph.D. in Management Science and Operations</i>                      | London, UK<br>2025 – 2030 (Expected) |
| <b>Sharif University of Technology</b><br><i>M.Sc. in Industrial Engineering (Optimization Methods)</i> | Tehran, Iran<br>2023 – 2025          |
| <b>Sharif University of Technology</b><br><i>B.Sc. in Industrial Engineering</i>                        | Tehran, Iran<br>2019 – 2023          |

## Research Interests

**Methodology:** Optimization, machine learning.

**Applications:** Healthcare operations, sustainable operations.

## Selected Publications

- [1] **Amani Bani, E., & Eshghi, K.** (2026). *Quantum modeling of the dynamic ride-sharing problem: Development of quantum Benders decomposition methods*. *Expert Systems with Applications*, 310, 131320. [DOI]
- [2] Fallahi, A., **Amani Bani, E.**, & Varmazyar, M. (2025). *Towards sustainable scheduling of unrelated parallel batch processors: A multi-objective approach with triple bottom line, classical and data-driven robust optimization*. *Computers & Operations Research*, 173, 106863. [DOI]
- [3] **Amani Bani, E.**, Fallahi, A., Varmazyar, M., & Fathi, M. (2022). *Designing a sustainable reverse supply chain network for COVID-19 vaccine waste under uncertainty*. *Computers & Industrial Engineering*, 174, 108808. [DOI]
- [4] Fallahi, A., **Amani Bani, E.**, & Niaki, S. T. A. (2022). *A constrained multi-item EOQ inventory model for reusable items: Reinforcement learning-based differential evolution and particle swarm optimization*. *Expert Systems with Applications*, 207, 118018. [DOI]

## Professional Experience

**Snapp!** Mar. 2025 – Sep. 2025  
*Senior AI Engineer* (Largest ride-hailing and delivery platform in the Middle East) *Tehran, Iran*

- Engineered low-latency AI pipelines for large-scale customer service operations.
- Built classification models to detect multihoming drivers, supporting platform-loyalty analytics.
- Designed ML-based behavioural prediction models informing dynamic incentive policies across a network of 2M+ drivers.

## FLYTZI

Co-founder & Optimization Engineer

Feb. 2024 – Feb. 2025

Tehran, Iran

- Co-founded a mid-mile logistics platform matching vendors with drivers for parcel delivery.
- Formulated and solved the underlying capacitated vehicle routing problem with time windows; developed metaheuristic solvers for real-time dispatch at operational scale.
- Designed a 3D bin-packing module for parcel-to-vehicle assignment, integrating geometric feasibility with routing decisions.

## Teaching Experience

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Sharif University of Technology, Department of Industrial Engineering

2022 – 2025

- **Head Teaching Assistant:** Statistics; Engineering Economics; Plant Layout; Project Control.
- **Teaching Assistant:** Operations Research; Probability Theory; Simulation.

## Academic Service

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**Journal Reviewer:** *Transportation Research Part E; Expert Systems with Applications; IEEE Transactions on Pattern Analysis and Machine Intelligence.*

## Volunteering

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**Gamein — Nationwide Gamified Education Event**

2022 – 2023

Game Design and Development Team Lead

Sharif University of Technology

- Led product development for a national educational simulation reaching 3,300+ participants across Iranian universities.
- Coordinated a cross-functional team spanning game design, software engineering, and operations modelling.

## Honors & Awards

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- LBS PhD Fellowship, full funding, London Business School, 2025–2030.
- Direct admission to M.Sc. programme (Talented Student Quota), Sharif University of Technology, 2023.
- Top 0.2% nationally, Iranian University Entrance Exam (Konkur), among 150,000+ candidates, 2019.

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Last updated: May, 2026